

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Staff Training Management System (STMS)

Project Duration: 6 Weeks
Date: July 2026

1. Introduction

1.1 Project Overview

The Staff Training Management System (STMS) is a web-based application designed to streamline and digitize the planning, scheduling, assignment, monitoring, and reporting of staff training programmes within the Nigerian Safety Investigation Bureau (NSIB).

The system provides a centralized platform for the Human Resources and Learning & Development Directorate to efficiently manage externally delivered training programmes across departments and regional offices. Rather than delivering training content, the platform focuses on the administrative processes surrounding staff development, ensuring that training activities are properly planned, coordinated, monitored, and documented.

The application will replace manual and fragmented training management processes with a secure, centralized, and scalable solution that improves operational efficiency, minimizes scheduling conflicts, and provides management with real-time visibility into organizational training activities.

1.2 Background

The Nigerian Safety Investigation Bureau (NSIB) is responsible for improving transportation safety through the investigation of aviation, rail, marine, and road transport accidents and incidents across Nigeria.

As part of its commitment to maintaining a highly skilled workforce, the Bureau organizes numerous professional training programmes annually for employees across multiple directorates and regional offices. These training programmes are conducted by external institutions, professional organizations, universities, consultants, and accredited training providers.

Currently, many aspects of the training administration process are managed manually using spreadsheets, documents, emails, and other disconnected tools. This approach makes it difficult to effectively coordinate training schedules, assign employees, monitor participation, manage certificates, and generate timely management reports.

The proposed Staff Training Management System will serve as a centralized web platform that digitizes these processes, improves visibility, and supports efficient decision-making while providing a scalable foundation for future enhancements.

1.3 Purpose of the Project

The purpose of this project is to develop a centralized web-based platform that enables the Nigerian Safety Investigation Bureau to efficiently manage the complete administrative lifecycle of staff training programmes.

The system will support the planning of annual training programmes, scheduling of training sessions, assignment of employees, monitoring of attendance and completion, management of training certificates, conflict detection, automated notifications, and generation of management reports.

The platform is intended to improve efficiency, reduce administrative workload, enhance compliance with organizational training requirements, and provide management with accurate, up-to-date information for decision-making.

1.4 Project Objectives

The project aims to achieve the following objectives:

- Develop a centralized web-based platform for managing staff training activities.
- Digitize the organization's training planning and scheduling processes.
- Enable HR and Learning & Development personnel to efficiently assign employees to training programmes.
- Prevent employee scheduling conflicts through automated conflict detection.
- Provide employees with a portal to view assigned training programmes and upload completion certificates.
- Improve monitoring of training participation and completion across departments and regions.

- Generate comprehensive reports that support management decision-making.
- Maintain a secure and scalable platform that supports future organizational growth and additional functionality.

1.5 Business Objectives

The implementation of the Staff Training Management System is expected to deliver the following business benefits:

- Centralize all organizational training information into a single platform.
- Improve planning and coordination of training programmes.
- Eliminate duplicate or conflicting employee training assignments.
- Increase visibility into staff training progress and compliance.
- Reduce reliance on manual spreadsheets and paper-based records.
- Improve reporting accuracy and reduce report preparation time.
- Support data-driven workforce development decisions.
- Establish a scalable foundation for future enhancements and integration with other organizational systems.

2. Project Scope

2.1 In Scope

The Staff Training Management System (STMS) will provide a centralized web-based platform for managing the administrative lifecycle of staff training programmes within the Nigerian Safety Investigation Bureau (NSIB). The system will support the following functional areas:

User Authentication and Access Control

- Secure user login and logout.
- Password management.
- Role-based access control.
- User profile management.

Organizational Structure Management

- Management of regions.
- Management of departments and directorates.
- Employee record management.
- Employee assignment to departments and regions.

Training Planning

- Creation and management of annual training plans.
- Management of training categories.
- Management of training priorities (High, Medium, Low).
- Management of training providers.
- Management of training venues.

Training Scheduling

- Scheduling of training programmes.
- Assignment of training dates and durations.
- Management of participant capacity.
- Scheduling across multiple regions and departments.

Employee Training Assignment

- Assignment of employees to training programmes.
- Department-wide training assignments.
- Multiple participant selection.
- Assignment history tracking.

Conflict Detection

- Automatic detection of employee scheduling conflicts.
- Prevention of assigning employees to overlapping training sessions.
- Display of conflict notifications before assignment is confirmed.

Training Monitoring

- Monitoring of training attendance.
- Monitoring of employee participation.
- Tracking of training completion status.
- Monitoring of mandatory training compliance.

Certificate Management

- Certificate upload by employees.
- PDF certificate storage.
- Certificate download.
- Certificate history.

Notifications

- Email notifications.
- In-app notifications.
- Training assignment notifications.
- Upcoming training reminders.
- Conflict notifications.
- Training completion notifications.

Reporting and Analytics

- Employee training reports.
- Department training reports.
- Regional training reports.
- Training calendar reports.
- Training completion reports.
- Compliance reports.
- Export of reports to PDF and Excel formats.

Administration

- User management.
- System configuration.
- Training management.
- Audit logging.
- Dashboard and analytics.

2.2 Out of Scope

The following features are outside the scope of Version 1 of the Staff Training Management System:

- Online learning or e-learning modules.
- Video conferencing or virtual classrooms.
- Online examinations or assessments.
- Payroll integration.
- Recruitment and onboarding management.
- Employee performance appraisal.
- Leave management.
- Financial accounting.
- Budget approval workflows.
- Mobile application.
- Third-party HR system integration.

These features may be considered as future enhancements based on organizational needs.

3. Stakeholders

The successful implementation of the Staff Training Management System involves multiple stakeholders within the organization.

Stakeholder	Responsibility
Executive Management	Project sponsorship, strategic oversight, approval of reports and performance monitoring.
Human Resources & Learning and Development	Planning, scheduling, assignment and monitoring of staff training programmes.
Department Heads	Nomination of staff for training programmes and monitoring departmental compliance.
Staff Members	Viewing assigned training, uploading certificates and monitoring personal training history.
System Administrator	User management, system configuration and maintenance.
Development Team	Design, development, testing, deployment and technical support of the web application.

4. User Roles

The system will support the following user roles.

Super Administrator

The Super Administrator has unrestricted access to the system and is responsible for overall administration.

Responsibilities include:

- Manage users.
- Manage roles and permissions.
- Configure system settings.
- Manage departments and regions.
- Access all reports.
- Monitor system activities.

HR / Learning & Development Administrator

This role manages all training-related activities.

Responsibilities include:

- Create annual training plans.
- Schedule training programmes.
- Assign employees to training.
- Manage training providers.
- Manage venues.
- Monitor attendance.
- Generate reports.
- Send notifications.

Department Head

Department Heads oversee staff development within their departments.

Responsibilities include:

- View departmental training plans.
- Nominate employees.
- Monitor staff participation.
- View departmental reports.

Staff

Employees interact with the system to manage their assigned training activities.

Responsibilities include:

- Login securely.
- View assigned training programmes.
- View training schedules.
- Receive notifications.
- Upload training certificates.
- View training history.
- Monitor completion status.

5. Business Objectives

The implementation of the Staff Training Management System seeks to achieve the following objectives:

- Digitize the organization's staff training management process.
- Improve coordination of training activities across all departments and regional offices.
- Eliminate scheduling conflicts involving employees.
- Provide centralized access to training records.
- Improve visibility into staff training participation and completion.
- Enhance reporting and management decision-making.
- Reduce administrative effort associated with manual record keeping.
- Improve compliance with mandatory organizational training programmes.
- Provide a scalable platform capable of supporting future organizational growth.

6. Functional Requirements

6.1 Authentication & User Management

Module Overview

The Authentication and User Management module provides secure access to the Staff Training Management System while ensuring that users can only access features and information appropriate to their assigned roles.

Functional Requirements

The system shall:

- Allow registered users to securely log in using their email address and password.
- Allow authenticated users to securely log out of the system.
- Support password reset functionality through registered email addresses.
- Enforce role-based access control for all users.
- Allow administrators to create, edit, activate and deactivate user accounts.
- Allow users to update their profile information.
- Maintain user session security throughout application usage.

6.2 Region Management

Module Overview

The Region Management module enables the organization to manage its regional offices and associate departments and employees with their respective locations.

Functional Requirements

The system shall:

- Allow administrators to create, edit and deactivate regions.
- Maintain information for all organizational regions.
- Associate departments with regions.
- Associate employees with regions.
- Support region-based reporting and filtering.

6.3 Department Management

Module Overview

This module manages organizational departments and directorates within the system.

Functional Requirements

The system shall:

- Create new departments.
- Edit department information.
- Assign department heads.
- Associate departments with regions.
- View department details.
- Archive inactive departments.

6.4 Employee Management

Module Overview

This module manages employee records and maintains staff information required for training administration.

Functional Requirements

The system shall:

- Register employees.
- Maintain employee profiles.
- Store employee ID.
- Store designation.
- Store department.
- Store region.
- Store employment status.
- Search employees.
- Filter employees by department, region or employment status.
- View employee training history.
- Maintain employee assignment records.

6.5 Training Provider Management

Module Overview

Training providers are external organizations responsible for delivering staff training programmes.

Functional Requirements

The system shall:

- Register training providers.
- Edit provider information.
- Archive inactive providers.
- Associate providers with training programmes.
- Maintain provider history.

6.6 Annual Training Planning

Module Overview

The Annual Training Planning module serves as the foundation of the Staff Training Management System. It enables the Human Resources and Learning & Development Directorate to prepare, organize and maintain the organization's annual training plan before individual training programmes are scheduled.

The annual training plan acts as the organization's official training roadmap and contains all approved training programmes intended for implementation within a training year.

Functional Requirements

The system shall allow authorized users to:

- Create annual training plans.
- Specify the training year.
- Add multiple planned training programmes to a training plan.
- Assign each training programme to a department or directorate.
- Specify the expected number of participants.
- Assign training providers.
- Specify training priorities.
- Record proposed venues.
- Record estimated budgets.
- Categorize training as local or foreign.
- Save training plans as drafts before approval.
- Update draft plans before implementation.
- View historical annual training plans.

Information Captured

Each planned training programme shall contain the following information:

- Training Title
- Directorate/Department
- Training Provider

- Training Type
- Priority Level
- Expected Participants
- Proposed Venue
- Estimated Budget
- Proposed Duration
- Training Description
- Training Year
- Current Status

6.7 Training Scheduling

Module Overview

Once an annual training plan has been approved, individual training programmes can be scheduled for implementation.

The Training Scheduling module allows administrators to assign actual dates and logistical details to approved training programmes while ensuring efficient coordination across departments and regions.

Functional Requirements

The system shall allow authorized users to:

- Select a training programme from the approved annual training plan.
- Assign a training start date.
- Assign a training end date.
- Specify training duration.
- Confirm training venue.
- Confirm training provider.
- Publish the training schedule.
- View scheduled training in calendar format.
- Filter schedules by department, region or month.
- Cancel scheduled training when necessary.
- Maintain scheduling history.

Business Rules

The system shall enforce the following rules:

- Only approved training plan entries may be scheduled.
- A training programme may only have one active schedule.
- Published schedules cannot be edited directly.
- Changes to published schedules shall require cancellation and rescheduling.
- Multiple training programmes may occur on the same day.
- Multiple training programmes may occur at the same time provided they involve different employees.

6.8 Employee Training Assignment

Module Overview

Following the scheduling of a training programme, employees may be nominated and assigned to participate. The assignment process ensures that only eligible employees are enrolled while preventing scheduling conflicts.

Functional Requirements

The system shall allow authorized users to:

- Assign individual employees.
- Assign multiple employees simultaneously.
- Assign entire departments.
- View assigned participants.
- Remove participants before commencement.
- Replace participants.
- Track assignment history.
- View assignment statistics.

Business Rules

The system shall enforce the following rules:

- Employees cannot be assigned to overlapping training schedules.
- Assignment shall not exceed the approved participant capacity.
- Employees may participate in multiple training programmes provided there is no scheduling conflict.
- Priority level shall guide assignment decisions where scheduling conflicts exist.
- Every assignment shall generate a notification to the employee.

6.9 Conflict Detection

Module Overview

The Conflict Detection module automatically validates employee assignments to prevent scheduling conflicts.

Functional Requirements

The system shall:

- Detect overlapping training assignments.
- Prevent assigning an employee to two training sessions occurring at the same time.
- Allow different employees to attend different simultaneous trainings.
- Display conflict warnings before assignments are confirmed.
- Record conflict logs for administrative review.

6.10 Attendance & Training Completion

Module Overview

This module monitors employee participation and completion of assigned training programmes.

Functional Requirements

The system shall:

- Record employee attendance.
- Mark completed training programmes.
- Display completion status.
- Maintain completion history.
- Generate completion statistics.

6.11 Certificate Management

Module Overview

Employees may upload certificates obtained after completing externally delivered training programmes.

Functional Requirements

The system shall:

- Accept PDF certificate uploads.
- Associate certificates with completed training programmes.
- Allow certificate downloads.
- Maintain certificate history.
- Permit multiple certificates where applicable.

6.12 Notification Management

Module Overview

The Notification module provides timely communication to users regarding training activities.

Functional Requirements

The system shall:

- Send training assignment notifications.
- Send upcoming training reminders.
- Notify users of scheduling conflicts.
- Notify users upon successful training completion.

- Support both in-app and email notifications.
- Maintain notification history.

6.13 Reporting & Analytics

Module Overview

The Reporting module provides management with comprehensive insights into organizational training activities.

Functional Requirements

The system shall:

- Generate employee training reports.
- Generate departmental training reports.
- Generate regional training reports.
- Generate annual training reports.
- Generate training completion reports.
- Generate compliance reports.
- Display training dashboards.
- Export reports to PDF.
- Export reports to Microsoft Excel.

6.14 Dashboard

Module Overview

The dashboard provides users with a summarized view of information relevant to their role.

Functional Requirements

The system shall display:

- Total employees.
- Total departments.
- Total regions.

- Total scheduled training.
- Upcoming training.
- Completed training.
- Pending training.
- Training completion rate.
- Notifications.
- Recent activities.

7. Business Rules

The Staff Training Management System shall enforce the following business rules to ensure consistency, accuracy, and compliance with the organization's training management processes.

7.1 Annual Training Planning

- Every training programme must belong to an Annual Training Plan.
- Each Annual Training Plan shall be associated with a specific training year.
- Training programmes shall be created before they can be scheduled.
- Draft training plans may be modified before approval.
- Historical training plans shall remain accessible for reporting purposes.

7.2 Training Scheduling

- Only approved training programmes may be scheduled.
- A training programme may only have one active schedule at a time.
- Multiple training programmes may occur on the same day.
- Multiple training programmes may occur simultaneously provided different employees are assigned.
- Published training schedules cannot be edited directly.
- Cancelled schedules may be rescheduled.

7.3 Employee Assignment

- Employees may be assigned individually or as part of a department.

- Employees may participate in multiple training programmes within a training year.
- Employees shall not be assigned to overlapping training schedules.
- The number of assigned participants shall not exceed the approved participant capacity.
- Every assignment shall be recorded for audit purposes.

7.4 Priority Levels

Each training programme shall be assigned one of the following priority levels:

- High Priority
- Medium Priority
- Low Priority

Priority shall be used to support scheduling decisions where conflicts arise.

7.5 Attendance & Completion

A training programme shall only be marked as completed when:

- Attendance has been confirmed.
- The participant has completed the training.
- Completion has been recorded in the system.

7.6 Certificate Management

- Certificates shall be uploaded by participants.
- Certificates shall be uploaded in PDF format.
- Multiple certificates may be attached where applicable.
- Uploaded certificates shall remain available for future reference.

7.7 Notifications

The system shall automatically notify users when:

- A training assignment is created.
- A training schedule is approaching.
- A scheduling conflict is detected.
- A training programme has been completed.

Notifications shall be delivered through both in-app alerts and email where applicable.

7.8 Reporting

Reports shall always reflect current system data and shall support export to PDF and Microsoft Excel.

8. System Workflow

The Staff Training Management System follows a structured workflow that mirrors the organization's training administration process.

Phase 1 – Annual Training Planning

The Human Resources and Learning & Development Directorate prepares the organization's Annual Training Plan by defining all proposed training programmes for the year.

Information captured includes:

- Training title
- Directorate
- Training provider
- Expected participants
- Priority level
- Venue
- Budget
- Training type

At this stage, no dates are assigned.

Phase 2 – Training Scheduling

Once a training programme has been approved, administrators assign operational details such as:

- Start date
- End date
- Duration
- Venue confirmation
- Provider confirmation

The schedule is then published.

Phase 3 – Employee Assignment

Following publication, eligible employees are assigned to scheduled training programmes.

Before an assignment is completed, the system validates:

- Employee availability.
- Existing assignments.
- Participant capacity.
- Scheduling conflicts.

Successful assignments automatically generate notifications.

Phase 4 – Training Participation

- Employees attend externally delivered training programmes.
- Training providers conduct the training outside the web application.
- The system only records participation.

Phase 5 – Certificate Submission

- After successfully completing training, participants upload their certificates through the system.
- Certificates become part of each employee's permanent training history.

Phase 6 – Monitoring & Reporting

HR and management monitor:

- Training participation.
- Completion rates.
- Department compliance.
- Regional performance.
- Upcoming schedules.
- Historical records.

Management reports can be generated at any time.

9. Success Criteria

The Staff Training Management System shall be considered successful when it achieves the following outcomes:

- Training records are centralized within a single web-based platform.
- Annual training planning is fully digitized.
- Training schedules are managed electronically.
- Employee scheduling conflicts are automatically prevented.
- Employees can access their assigned training programmes online.
- Certificates are stored electronically.
- Management reports are generated efficiently.
- Administrative effort associated with manual record keeping is significantly reduced.
- The system provides a secure, reliable, and scalable platform for future organizational needs.

10. Assumptions

The following assumptions have been made during the preparation of this Product Requirements Document:

- The organization maintains accurate employee records.

- Each employee has a unique identifier.
- Employees have valid email addresses for notifications.
- Training programmes are conducted externally by approved providers.
- Users will access the system using modern web browsers.
- Internet connectivity is available across all regional offices.

11. Constraints

The implementation of the Staff Training Management System shall be subject to the following constraints:

- The project implementation period is six (6) weeks.
- The application shall be web-based.
- The system shall not deliver training content.
- The initial implementation shall support the organization's existing training processes without requiring major process changes.
- Future enhancements shall be accommodated through modular system design.

12. Future Enhancements

The proposed system has been designed with scalability in mind. Future versions may include:

- Training request and approval workflow.
- Training budget management.
- Vendor and provider portal.
- Mobile application.
- Integration with external HR systems.
- Calendar synchronization.
- Advanced analytics and dashboards.

- AI-assisted training recommendations.
- Competency and skills management.
- API integration with third-party systems.

13. Product Summary

The Staff Training Management System (STMS) is a centralized web-based application developed to streamline the planning, scheduling, assignment, monitoring, and reporting of staff training programmes within the Nigerian Safety Investigation Bureau.

The system focuses on managing the administrative aspects of externally delivered training programmes, enabling Human Resources and Learning & Development personnel to coordinate organizational training activities efficiently while providing management with accurate, real-time information for strategic decision-making.

By replacing manual processes with a secure, structured, and scalable platform, the STMS will improve operational efficiency, reduce scheduling conflicts, enhance compliance, and provide a reliable digital record of staff training across all departments and regional offices.

14. Feature Summary Matrix

The table below provides a high-level overview of the major functional modules included in the Staff Training Management System.

Module	Primary Users	Purpose
Authentication & User Management	All Users	Provides secure access to the system through role-based authentication and user account management.
Region Management	Super Administrator	Manages the organization's regional offices and associates departments and employees with their respective regions.
Department Management	Super Administrator	Maintains organizational departments and directorates.
Employee Management	HR / Learning & Development	Maintains employee records and organizational information required for training administration.

Module	Primary Users	Purpose
Training Provider Management	HR / Learning & Development	Manages external organizations responsible for delivering training programmes.
Annual Training Planning	HR / Learning & Development	Creates and manages the organization's annual training plan before implementation.
Training Scheduling	HR / Learning & Development	Schedules approved training programmes by assigning dates, venues and providers.
Employee Training Assignment	HR / Learning & Development	Assigns employees or departments to scheduled training programmes.
Conflict Detection	System	Automatically prevents employees from being assigned to overlapping training schedules.
Attendance & Completion	HR / Learning & Development	Records attendance and monitors employee training completion.
Certificate Management	Staff	Enables employees to upload and manage training certificates.
Notification Management	System	Sends training assignments, reminders and system notifications via email and in-app alerts.
Reports & Analytics	Management / HR	Generates organizational reports, dashboards and compliance statistics.
Dashboard	All Users	Provides role-based summaries and key performance indicators relevant to each user.

Appendix

Glossary

- STMS: Staff Training Management System
- NSIB: Nigerian Safety Investigation Bureau
- HR: Human Resources
- L&D: Learning and Development
- Training Programme: An approved staff development activity planned by the organization.
- Training Schedule: The confirmed date, venue and logistics for a training programme.
- Training Assignment: Allocation of employees to a scheduled training programme.
- Training Provider: An external organization responsible for delivering training.
- Conflict Detection: The process of identifying overlapping employee training schedules.